

LEON AP, Spain

WMO: 080550

Lat: 42.588N

Lon: 5.651W

Elev: 916

StdP: 90.79

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	-3.4	-9.6	1.8	1.5	-7.3	2.3	1.5	12.2	6.1	11.0	7.6	1.2	330	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	32.3	17.1	30.7	16.7	29.0	16.4	18.7	28.3	17.9	27.5	17.2	26.5	3.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	12.3	21.4	14.7	11.6	20.4	13.9	11.1	20.0	57.3	28.1	54.3	27.8	51.9	26.7	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.2	7.8	DB	-8.5	35.7	1.9	1.0	-9.9	36.4	-11.0	37.0	-12.1	37.6	-13.6	38.4	(4)
(5)			WB	-9.0	20.7	1.9	2.5	-10.4	22.5	-11.5	24.0	-12.5	25.4	-13.9	27.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	3.3	4.7	7.3	9.2	12.9	17.2	20.1	19.7	17.1	12.7	7.4	4.9	(6)	
	DBStd	6.64	2.77	3.22	3.23	3.52	3.00	4.09	2.98	3.04	3.46	3.20	3.23	2.60	(7)	
	HDD10.0	774	207	149	96	56	7	1	0	0	0	12	88	159	(8)	
	HDD18.3	2723	466	382	343	273	169	71	17	20	65	174	327	417	(9)	
	CDD10.0	1293	0	1	11	33	98	216	313	301	212	97	11	1	(10)	
	CDD18.3	199	0	0	0	0	2	36	72	63	27	1	0	0	(11)	
	CDH23.3	2798	0	0	0	5	63	500	993	866	343	29	0	0	(12)	
	CDH26.7	961	0	0	0	0	9	169	362	312	107	2	0	0	(13)	
(14) Wind	WSAvg	3.0	2.8	3.5	3.8	3.7	3.1	2.8	2.7	2.7	2.6	2.5	2.8	2.4	(14)	
(15) Precipitation	PrecAvg	570	62	56	34	49	52	38	25	21	41	61	63	67	(15)	
	PrecMax	799	253	188	155	139	115	120	110	84	143	159	166	242	(16)	
	PrecMin	363	0	5	0	3	4	3	0	0	0	4	0	4	(17)	
	PrecStd	114	50	46	30	37	27	27	26	22	40	39	44	61	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	16.7	21.5	24.0	28.0	33.5	33.8	34.1	32.7	25.7	18.5	15.0	(19)	
		MCWB	7.2	8.7	10.5	11.4	14.3	16.9	17.1	17.6	17.2	12.9	11.7	9.1	(20)	
	2%	DB	11.4	14.1	19.1	21.7	25.0	30.8	32.2	31.9	29.8	23.9	16.1	12.2	(21)	
		MCWB	7.4	8.1	9.8	11.1	13.1	17.0	17.2	16.7	16.8	13.2	10.7	8.8	(22)	
	5%	DB	9.8	11.9	17.0	19.2	23.0	28.5	30.6	30.0	27.1	21.8	14.2	10.7	(23)	
		MCWB	6.6	7.5	9.4	10.9	13.0	16.3	16.6	16.5	15.6	13.0	10.3	8.1	(24)	
	10%	DB	8.2	10.0	14.6	16.8	20.8	26.2	28.9	28.3	24.9	19.2	12.4	9.2	(25)	
		MCWB	6.0	6.9	8.8	10.2	12.0	15.6	16.6	16.0	14.7	12.3	9.7	7.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	10.7	12.1	13.9	15.7	19.1	20.5	19.7	18.4	16.1	13.5	11.2	(27)	
		MCDB	11.7	13.5	18.9	20.9	23.5	27.8	30.2	31.0	28.8	21.0	16.4	12.1	(28)	
	2%	WB	8.7	9.3	10.9	12.5	14.7	18.2	19.0	18.5	17.1	14.8	12.1	9.6	(29)	
		MCDB	10.1	12.3	17.0	18.6	22.4	27.6	28.0	28.7	27.0	20.3	14.7	10.9	(30)	
	5%	WB	7.3	8.2	10.0	11.5	13.7	17.3	18.0	17.5	16.3	13.9	11.2	8.6	(31)	
		MCDB	8.8	10.9	15.6	17.4	20.5	26.5	27.3	27.2	25.0	19.5	13.2	9.9	(32)	
	10%	WB	6.2	7.3	8.9	10.6	12.9	16.2	17.2	16.7	15.6	13.0	10.1	7.6	(33)	
		MCDB	7.8	9.8	13.5	16.3	19.0	24.7	26.5	25.7	23.1	17.6	12.1	8.9	(34)	
(35) Mean Daily Temperature Range	MDBR		8.4	8.9	11.5	11.3	12.9	13.7	15.3	15.5	13.5	10.9	8.3	7.9	(35)	
	5% DB	MCDBR	11.6	12.3	16.4	16.7	17.6	17.1	18.0	17.6	16.5	15.7	11.4	11.2	(36)	
		MCWBR	8.0	7.6	9.0	8.0	7.9	6.1	6.0	6.2	6.4	7.2	7.1	7.7	(37)	
	5% WB	MCDBR	8.0	10.4	13.1	13.8	13.8	14.7	15.3	15.0	14.4	11.8	8.9	7.4	(38)	
		MCWBR	6.2	6.9	7.6	7.1	6.6	5.7	6.1	6.3	6.3	6.2	6.0	5.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.281	0.292	0.324	0.337	0.351	0.356	0.343	0.346	0.330	0.319	0.292	0.278	(40)	
	taud		2.534	2.503	2.431	2.402	2.380	2.384	2.403	2.393	2.449	2.500	2.541	2.540	(41)	
	Ebn at Noon		876	921	923	931	921	914	922	910	903	874	854	848	(42)	
	Edh at Noon		69	83	101	112	118	118	115	112	98	83	68	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.73	2.69	3.87	5.09	6.17	6.99	7.23	6.39	4.91	3.07	1.93	1.53	(44)	
	RadStd		0.21	0.31	0.53	0.57	0.52	0.39	0.34	0.33	0.25	0.34	0.28	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page